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Applied Calculus IV

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Week 7

■ 4.3

■ 4.4

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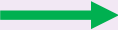
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Higher-Order Differential Equations

- 4.1 Preliminary Theory—Linear Equations
- 4.2 Reduction of Order
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- 4.4 Undetermined Coefficients—Superposition Approach
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- 4.9 Solving Systems of Linear DEs by Elimination
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Homogeneous Linear Differential Equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

A simpler version

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

with Constant Coefficients

Or
$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0$$

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0$$

Define $y = e^{rx}$

Thus, $y' = r e^{rx}$

$$y'' = r^2 e^{rx}$$

$$y''' = r^3 e^{rx}$$

$\vdots$

$$y^{(n)} = r^n e^{rx}$$

Replace

$$e^{rx} (a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0) = 0$$

$$e^{rx} \neq 0$$

$$a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = 0$$

namely characteristic equation

First-order DE

$$(a_1 \neq 0) \quad a_1 y' + a_0 y = 0$$

Define $y = e^{rx}$

$$\rightarrow e^{rx} (a_1 r + a_0) = 0 \quad \longrightarrow \quad a_1 r + a_0 = 0 \quad \longrightarrow \quad r = -\frac{a_0}{a_1} \quad \longrightarrow \quad y = ce^{-\frac{a_0}{a_1}x}$$

Example $y' + 2y = 0$

Define $y = e^{rx} \quad \longrightarrow \quad r + 2 = 0 \quad \longrightarrow \quad r = -2 \quad \longrightarrow \quad y = ce^{-2x}$

Second-order DE

Example $2y'' - 5y' - 3y = 0$

Define $y = e^{rx} \longrightarrow 2r^2 - 5r - 3 = 0$

$\longrightarrow (2r + 1)(r - 3) = 0 \longrightarrow r_1 = -\frac{1}{2} \text{ and } r_2 = 3$

$\longrightarrow e^{-\frac{1}{2}x} \text{ and } e^{3x} \text{ are fundamental solutions}$

$\longrightarrow y = c_1 e^{-\frac{1}{2}x} + c_2 e^{3x}$

Second-order DE

Example $y'' - 10y' + 25y = 0$

Define $y = e^{rx} \longrightarrow r^2 - 10r + 25 = 0 \longrightarrow (r - 5)^2 = 0$

$\longrightarrow r = 5$ Repeated root

$\longrightarrow e^{5x}$ is a fundamental solution

$\longrightarrow xe^{5x}$ is another fundamental solution

See 4.2

$$y_2 = y_1(x) \int \frac{e^{-\int P(x)dx}}{y_1^2(x)} dx$$

$$y_2 = e^{5x} \int \frac{e^{-10x}}{e^{10x}} dx$$

$\longrightarrow y = (c_1 + c_2x)e^{5x}$

Second-order DE

Example $y'' + 4y' + 5y = 0$

Define $y = e^{rx} \longrightarrow r^2 + 4r + 5 = 0$

$\longrightarrow (r + 2)^2 = -1 = i^2 \longrightarrow r = -2 \pm i$ complex conjugate roots

$\longrightarrow e^{-2x}\cos x$ and $e^{-2x}\sin x$ are fundamental solutions (why?)

$\longrightarrow y = e^{-2x} (c_1 \cos(x) + c_2 \sin(x))$

Remark:

$e^{(-2+i)x}$ and $e^{(-2-i)x}$ are fundamental solutions $\rightarrow y = c_1 e^{(-2+i)x} + c_2 e^{(-2-i)x}$ is a general solution

We know that $e^{ix} = \cos x + i \sin x$

$$\begin{aligned} e^{(-2+i)x} &= e^{-2x}(\cos x + i \sin x) \\ e^{(-2-i)x} &= e^{-2x}(\cos x - i \sin x) \end{aligned}$$

Replace

$$\rightarrow y = \overbrace{(c_1 + c_2)}^{C_1} e^{-2x}(\cos x) + i \overbrace{(c_1 - c_2)}^{C_2} e^{-2x}(\sin x)$$

$$y = C_1 e^{-2x}(\cos x) + C_2 e^{-2x}(\sin x)$$

Second-order DE

Example $y'' + 4y' + 7y = 0$

Define $y = e^{rx} \rightarrow r^2 + 4r + 7 = 0$

$\rightarrow (r + 2)^2 = -3 = 3i^2 \rightarrow r = -2 \pm \sqrt{3}i$ complex conjugate roots

$\rightarrow e^{-2x}\cos\sqrt{3}x$ and $e^{-2x}\sin\sqrt{3}x$ are fundamental solutions

$\rightarrow y = e^{-2x} (c_1 \cos(\sqrt{3}x) + c_2 \sin(\sqrt{3}x))$

$r = \alpha \pm \beta i \rightarrow y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$

Summary (Quadratic DE)

$$ay'' + by' + cy = 0,$$

Define $y = e^{rx} \longrightarrow ar^2 + br + c = 0$

Two distinct real roots r_1 and r_2 $y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$

One repeated real root r $y = (c_1 + c_2 x) e^{rx}$

Complex conjugate roots $r = \alpha \pm \beta i$ $y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$

Example: Solve $4y'' + 4y' + 17y = 0$, $y(0) = 1, y'(0) = 2$

Solution:

Define $y = e^{rx} \rightarrow 4r^2 + 4r + 17 = 0$

$$4\left(r^2 + r + \frac{1}{4}\right) = -16 \rightarrow 4\left(r + \frac{1}{2}\right)^2 = -16 \rightarrow \left(r + \frac{1}{2}\right)^2 = 4i^2 \rightarrow \left(r + \frac{1}{2}\right) = \pm 2i \rightarrow r = -\frac{1}{2} \pm 2i$$

Complex conjugate roots

$$\rightarrow y(x) = e^{-\frac{1}{2}x} (c_1 \cos 2x + c_2 \sin 2x)$$

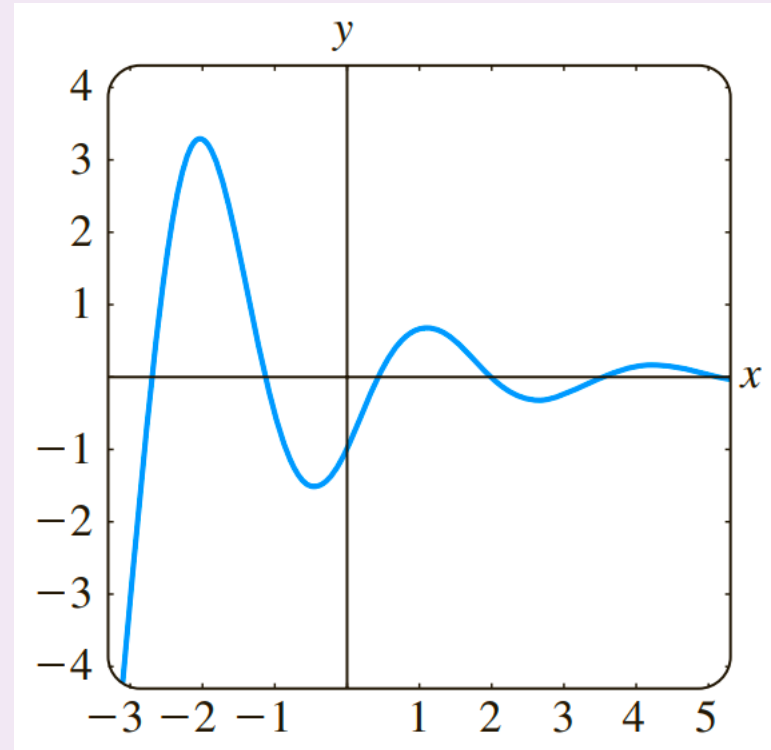
Apply conditions $y(0) = e^0 (c_1 \cos(0) + c_2 \sin(0)) = c_1 = -1$

$$y'(0) = e^0 \left(\left(\frac{1}{2} + 2c_2\right) \cos(0) + \left(2 - \frac{c_2}{2}\right) \sin(0) \right) = \frac{1}{2} + 2c_2 = 2 \rightarrow c_2 = \frac{3}{4}$$

➡ $y = e^{-\frac{1}{2}x} \left(-\cos(2x) + \frac{3}{4} \sin(2x) \right)$

✓ y is oscillatory

✓ $x \rightarrow \infty \Rightarrow y \rightarrow 0$



Example: Solve $y''' + 3y'' - 4y = 0$.

Solution:

Define $y = e^{rx} \longrightarrow r^3 + 3r^2 - 4 = 0 \longrightarrow (r - 1)(r + 2)^2 = 0$

e^x is a fundamental solution

e^{-2x} and xe^{-2x} are fundamental solution

$\longrightarrow y = c_1 e^x + c_2 e^{-2x} + c_3 x e^{-2x}$

Example: Solve $\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0$.

Solution:

Define $y = e^{rx} \longrightarrow r^4 + 2r^2 + 1 = 0 \longrightarrow (r^2 + 1)^2 = 0 \longrightarrow r = \pm i$

repeated - complex conjugate roots

$\cos x$ and $\sin x$ are fundamental solution

$x \cos x$ and $x \sin x$ are fundamental solution



$$y = (c_1 + c_2x) \cos x + (c_3 + c_4x) \sin x$$

Example: Solve $3y''' + 5y'' + 10y' - 4y = 0$

Solution:

Define $y = e^{rx} \rightarrow 3r^3 + 5r^2 + 10r - 4 = 0$

$\rightarrow (3r - 1)(r^2 + 2r + 4) = 0$

$\rightarrow r = \frac{1}{3} \text{ \& } -1 \pm \sqrt{3}i$

$e^{\frac{1}{3}x}$ is a fundamental solution

complex conjugate roots

$e^{-x} \cos x$ and $e^{-x} \sin x$ are fundamental solution



$$y = c_1 e^{\frac{1}{3}x} + c_2 e^{-x} \cos(\sqrt{3}x) + c_3 e^{-x} \sin(\sqrt{3}x)$$

Exercises 4.3

In Problems 1–14 find the general solution of the given second-order differential equation.

1. $4y'' + y' = 0$

2. $y'' - 36y = 0$

3. $y'' - y' - 6y = 0$

4. $y'' - 3y' + 2y = 0$

5. $y'' + 8y' + 16y = 0$

6. $y'' - 10y' + 25y = 0$

7. $12y'' - 5y' - 2y = 0$

8. $y'' + 4y' - y = 0$

9. $y'' + 9y = 0$

10. $3y'' + y = 0$

11. $y'' - 4y' + 5y = 0$

12. $2y'' + 2y' + y = 0$

13. $3y'' + 2y' + y = 0$

14. $2y'' - 3y' + 4y = 0$

Solution

1. $4y'' + y' = 0$

Define $y = e^{rx} \longrightarrow 4r^2 + r = 0$

$\longrightarrow r = 0 \text{ \& } -\frac{1}{4}$

$\longrightarrow y = c_1 e^{0 \cdot x} + c_2 e^{-\frac{1}{4}x}$

$\longrightarrow y = c_1 + c_2 e^{-\frac{1}{4}x}$

In Problems 15–28 find the general solution of the given higher-order differential equation.

15. $y''' - 4y'' - 5y' = 0$

16. $y''' - y = 0$

17. $y''' - 5y'' + 3y' + 9y = 0$

18. $y''' + 3y'' - 4y' - 12y = 0$

19. $\frac{d^3u}{dt^3} + \frac{d^2u}{dt^2} - 2u = 0$

20. $\frac{d^3x}{dt^3} - \frac{d^2x}{dt^2} - 4x = 0$

21. $y''' + 3y'' + 3y' + y = 0$

22. $y''' - 6y'' + 12y' - 8y = 0$

23. $y^{(4)} + y''' + y'' = 0$

24. $y^{(4)} - 2y'' + y = 0$

25. $16 \frac{d^4y}{dx^4} + 24 \frac{d^2y}{dx^2} + 9y = 0$

26. $\frac{d^4y}{dx^4} - 7 \frac{d^2y}{dx^2} - 18y = 0$

27. $\frac{d^5u}{dr^5} + 5 \frac{d^4u}{dr^4} - 2 \frac{d^3u}{dr^3} - 10 \frac{d^2u}{dr^2} + \frac{du}{dr} + 5u = 0$

28. $2 \frac{d^5x}{ds^5} - 7 \frac{d^4x}{ds^4} + 12 \frac{d^3x}{ds^3} + 8 \frac{d^2x}{ds^2} = 0$

Solution

15. $y''' - 4y'' - 5y' = 0$

Define $y = e^{rx} \longrightarrow r^3 - 4r^2 - 5r = 0 \longrightarrow r(r - 5)(r + 1) = 0$

$\longrightarrow r = 0, \quad 5 \quad \& \quad -1$

$\longrightarrow e^{0x}, e^{5x}$ and e^{-x} are fundamental solutions

$\longrightarrow y = c_1 + c_2 e^{5x} + c_3 e^{-x}$

Solution: 16. $y''' - y = 0$

Define $y = e^{rx} \longrightarrow r^3 - 1 = 0 \longrightarrow (r - 1)(r^2 + r + 1) = 0$

$\longrightarrow r = 1, \quad -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$

$\longrightarrow e^x$, $e^{-\frac{1}{2}x} \cos \frac{\sqrt{3}}{2}x$ and $e^{-\frac{1}{2}x} \sin \frac{\sqrt{3}}{2}x$ are fundamental solutions

$\longrightarrow y = c_1 e^x + c_2 e^{-\frac{1}{2}x} \cos \left(\frac{\sqrt{3}}{2}x \right) + c_3 e^{-\frac{1}{2}x} \sin \left(\frac{\sqrt{3}}{2}x \right)$

In Problems 29–36 solve the given initial-value problem.

29. $y'' + 16y = 0, \quad y(0) = 2, y'(0) = -2$

30. $\frac{d^2y}{d\theta^2} + y = 0, \quad y(\pi/3) = 0, y'(\pi/3) = 2$

31. $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} - 5y = 0, \quad y(1) = 0, y'(1) = 2$

32. $4y'' - 4y' - 3y = 0, \quad y(0) = 1, y'(0) = 5$

33. $y'' + y' + 2y = 0, \quad y(0) = y'(0) = 0$

34. $y'' - 2y' + y = 0, \quad y(0) = 5, y'(0) = 10$

35. $y''' + 12y'' + 36y' = 0, \quad y(0) = 0, y'(0) = 1, y''(0) = -7$

36. $y''' + 2y'' - 5y' - 6y = 0, \quad y(0) = y'(0) = 0, y''(0) = 1$

In Problems 37–40 solve the given boundary-value problem.

37. $y'' - 10y' + 25y = 0, \quad y(0) = 1, y(1) = 0$

38. $y'' + 4y = 0, \quad y(0) = 0, y(\pi) = 0$

39. $y'' + y = 0, \quad y'(0) = 0, y'(\pi/2) = 0$

40. $y'' - 2y' + 2y = 0, \quad y(0) = 1, y(\pi) = 1$

In Problems 41 and 42 solve the given problem, find the general solution in the following forms

$$y = c_1 e^{kx} + c_2 e^{-kx}.$$

$$y = c_1 \cosh kx + c_2 \sinh kx.$$

41. $y'' - 3y = 0, \quad y(0) = 1, y'(0) = 5$

42. $y'' - y = 0, \quad y(0) = 1, y'(1) = 0$

Recall

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

In Problems 43–48 each figure represents the graph of a particular solution of one of the following differential equations:

Match a solution curve with one of the differential equations.

Explain your reasoning.

(a) $y'' - 3y' - 4y = 0$

(c) $y'' + 2y' + y = 0$

(e) $y'' + 2y' + 2y = 0$

(b) $y'' + 4y = 0$

(d) $y'' + y = 0$

(f) $y'' - 3y' + 2y = 0$

43.

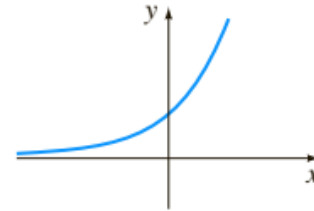


FIGURE 4.3.2 Graph for Problem 43

44.

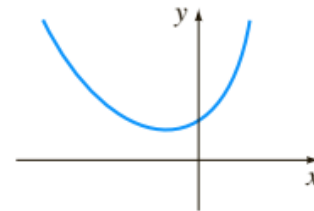


FIGURE 4.3.3 Graph for Problem 44

45.

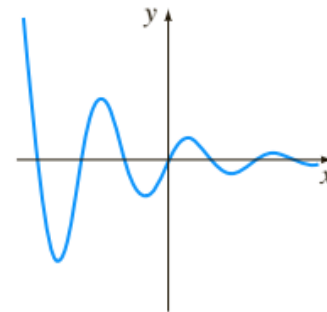


FIGURE 4.3.4 Graph for Problem 45

46.

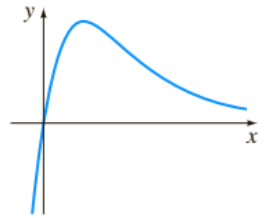


FIGURE 4.3.5 Graph for Problem 46

47.

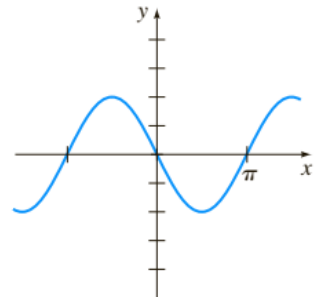


FIGURE 4.3.6 Graph for Problem 47

48.

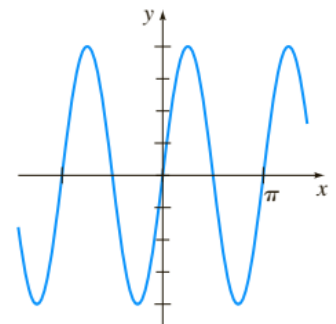


FIGURE 4.3.7 Graph for Problem 48

(a) $y'' - 3y' - 4y = 0$

Define $y = e^{rx} \longrightarrow r^2 - 3r - 4 = (r - 4)(r + 1) = 0 \longrightarrow r = 4 \text{ \& } -1$

$\longrightarrow y = c_1 e^{4x} + c_2 e^{-x}$

$x \rightarrow \infty \Rightarrow y \rightarrow \infty$

$x \rightarrow -\infty \Rightarrow y \rightarrow \infty$

y is not oscillatory



Fig 44

(b) $y'' + 4y = 0$

Define $y = e^{rx} \rightarrow r^2 + 4 = 0 \rightarrow r = \pm 2i$

$\rightarrow y = c_1 \cos(2x) + c_2 \sin(2x)$

y is oscillatory; the period is π



Fig 48

(In Fig 47, the period is 2π and it can't be a solution)

In Problems 49–58 find a homogeneous linear differential equation with constant coefficients whose general solution is given.

49. $y = c_1 e^x + c_2 e^{5x}$

50. $y = c_1 e^{-4x} + c_2 e^{-3x}$

51. $y = c_1 + c_2 e^{2x}$

52. $y = c_1 e^{10x} + c_2 x e^{10x}$

53. $y = c_1 \cos 3x + c_2 \sin 3x$

54. $y = c_1 \cosh 7x + c_2 \sinh 7x$

55. $y = c_1 e^{-x} \cos x + c_2 e^{-x} \sin x$

56. $y = c_1 + c_2 e^{2x} \cos 5x + c_3 e^{2x} \sin 5x$

57. $y = c_1 + c_2 x + c_3 e^{8x}$

58. $y = c_1 \cos x + c_2 \sin x + c_3 \cos 2x + c_4 \sin 2x$

Recall

Two distinct real roots r_1 and r_2 $y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$

One repeated real root r $y = (c_1 + c_2 x) e^{rx}$

Complex conjugate roots $r = \alpha \pm \beta i$ $y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$

Solution

$$49. (r - 1)(r - 5) = r^2 - 6r + 5 = 0 \rightarrow y'' - 6y' + 5y = 0$$

$$50. (r + 4)(r + 3) = r^2 + 7r + 12 = 0 \rightarrow y'' + 7y' + 12y = 0$$

$$51. (r - 0)(r - 2) = r^2 - 2r = 0 \rightarrow y'' - 2y' = 0$$

$$52. (r - 10)(r - 10) = r^2 - 20r + 100 = 0 \rightarrow y'' - 20y' + 100y = 0$$

Higher-Order Differential Equations

- 4.1 Preliminary Theory—Linear Equations
- 4.2 Reduction of Order
- 4.3 Homogeneous Linear Equations with Constant Coefficients
- 4.4 Undetermined Coefficients—Superposition Approach
- 4.5 Undetermined Coefficients—Annihilator Approach
- 4.6 Variation of Parameters
- 4.7 Cauchy-Euler Equations
- 4.8 Green's Functions
- 4.9 Solving Systems of Linear DEs by Elimination
- 4.10 Nonlinear Differential Equations

Nonhomogeneous Linear Differential Equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0$$

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x),$$



We develop the method for non-homogeneous version

Recall

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x),$$

$$y = y_c + y_p$$

The general solution of a nonhomogeneous equation = complementary function + any particular solution



general solution of the homogenous version

Challenge: how could we find a particular solution?

Undetermined Coefficients:

If $g(x)$ is a constant k , a polynomial function, an exponential function, a sine or cosine function, or finite sums and products of these functions.

Note that the method is limited only for some functions; and it is not applicable for some $g(x)$, for instance

$$g(x) = \ln x, \quad g(x) = \frac{1}{x}, \quad g(x) = \tan x, \quad g(x) = \sin^{-1}x,$$

Example: Solve $y'' + 4y' - 2y = 2x^2 - 3x + 6$.

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 + 4r - 2 = 0, \rightarrow$

$$\begin{aligned} r &= \frac{-4 \pm \sqrt{4^2 - 4 \times 1 \times (-2)}}{2 \times 1} \\ &= \frac{-4 \pm \sqrt{16 + 8}}{2} \\ &= \frac{-4 \pm \sqrt{24}}{2} \\ &= \frac{-4 \pm 2\sqrt{6}}{2} \\ &= -2 \pm \sqrt{6} \end{aligned}$$

$e^{(-2+\sqrt{6})x}$ and $e^{(-2-\sqrt{6})x}$ are fundamental solutions

 $y_c = c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x}$

Step2: Finding a particular solution of non-homogenous version

Define $y_p = Ax^2 + Bx + C$. $\longrightarrow y'_p = 2Ax + B$ and $y''_p = 2A$

Replace $y''_p + 4y'_p - 2y_p = 2A + 8Ax + 4B - 2Ax^2 - 2Bx - 2C = 2x^2 - 3x + 6$.

$$\longrightarrow -2A = 2, \quad 8A - 2B = -3, \quad 2A + 4B - 2C = 6.$$

$$\longrightarrow A = -1, B = -\frac{5}{2}, \text{ and } C = -9.$$

$$\longrightarrow y_p = -x^2 - \frac{5}{2}x - 9.$$

Step3: $y = y_c + y_p$

Example: Solve $y'' - y' + y = 2 \sin 3x$

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 - r + 1 = 0. \rightarrow r = \frac{1 \pm \sqrt{(-1)^2 - 4 \times 1 \times 1}}{2 \times 1}$

$$= \frac{1 \pm \sqrt{1 - 4}}{2}$$
$$= \frac{1 \pm \sqrt{-3}}{2}$$
$$= \frac{1 \pm i\sqrt{3}}{2}$$

$e^{\frac{1}{2}x} \cos \frac{\sqrt{3}}{2}x$ and $e^{\frac{1}{2}x} \sin \frac{\sqrt{3}}{2}x$ are
fundamental solutions

$\rightarrow y_c = e^{\frac{1}{2}x} \left(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x \right)$

Step2: Finding a particular solution of non-homogenous version

Define $y_p = A \cos 3x + B \sin 3x.$

→ $y'_p = -3A \sin 3x + 3B \cos 3x$

$$y''_p = -9A \cos 3x - 9B \sin 3x$$

Replace $y''_p - y'_p + y_p = (-8A - 3B) \cos 3x + (3A - 8B) \sin 3x = 2 \sin 3x$

→ $-8A - 3B = 0, \quad 3A - 8B = 2, \quad \rightarrow A = \frac{6}{73} \text{ and } B = -\frac{16}{73}$

→ $y_p = \frac{6}{73} \cos 3x - \frac{16}{73} \sin 3x.$

Step3: $y = y_c + y_p$

Example: Solve $y'' - y' + y = 5e^{2x}$.

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 - r + 1 = 0. \rightarrow r = \frac{1 \pm \sqrt{(-1)^2 - 4 \times 1 \times 1}}{2 \times 1}$

$$= \frac{1 \pm \sqrt{1 - 4}}{2}$$
$$= \frac{1 \pm \sqrt{-3}}{2}$$
$$= \frac{1 \pm i\sqrt{3}}{2}$$

$e^{\frac{1}{2}x} \cos \frac{\sqrt{3}}{2}x$ and $e^{\frac{1}{2}x} \sin \frac{\sqrt{3}}{2}x$ are
fundamental solutions

$\rightarrow y_c = e^{\frac{1}{2}x} \left(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x \right)$

Step2: Finding a particular solution of non-homogenous version

Define $y_p = Ae^{2x}$

→ $y'_p = 2Ae^{2x}$

$$y''_p = 4Ae^{2x}$$

Replace in $y'' - y' + y = 5e^{2x}$

$$\begin{aligned}y''_p - y'_p + y_p &= 4Ae^{2x} - 2Ae^{2x} + Ae^{2x} \\&= (4A - 2A + A)e^{2x} \\&= 3Ae^{2x}\end{aligned}$$

→ $A = \frac{5}{3}$ → $y_p = \frac{5}{3}e^{2x}$

Step3: $y = y_c + y_p$

Example: Solve $y'' - 2y' - 3y = 4x - 5 + 6xe^{2x}$

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 - 2r - 3 = 0 \rightarrow (r - 3)(r + 1) = 0$

e^{-x} and e^{3x} are fundamental solutions

$\Rightarrow y_c = c_1e^{-x} + c_2e^{3x}$

Step2: Finding a particular solution of non-homogenous version

Define $y_p = Ax + B + Cxe^{2x} + Ee^{2x}$

→ $y'_p = A + Ce^{2x} + 2Cxe^{2x} + 2Ee^{2x}$

$$y''_p = 4Ce^{2x} + 4Cxe^{2x} + 4Ee^{2x}$$

Replace in $y'' - 2y' - 3y = 4x - 5 + 6xe^{2x}$

$$-3Ax - 2A - 3B - 3Cxe^{2x} + (2C - 3E)e^{2x} = 4x - 5 + 6xe^{2x}$$

→ $-3A = 4, \quad -2A - 3B = -5, \quad -3C = 6, \quad 2C - 3E = 0.$

→ $y_p = -\frac{4}{3}x + \frac{23}{9} - 2xe^{2x} - \frac{4}{3}e^{2x}$

$$A = -\frac{4}{3}$$

$$B = \frac{23}{9}$$

$$C = -2$$

$$E = -\frac{4}{3}$$

Step3: $y = y_c + y_p$

TABLE 4.4.1 Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$

Example:

$$y'' - 8y' + 25y = 5x^3e^{-x} - 7e^{-x} \quad \Rightarrow \quad y_p = (Ax^3 + Bx^2 + Cx + E)e^{-x}$$

$$y'' + 4y = x \cos x \quad \Rightarrow \quad y_p = (Ax + B) \cos x + (Cx + E) \sin x.$$

Remark: Using the Superposition Principle

$$y'' - 9y' + 14y = 3x^2 - 5 \sin 2x + 8xe^{6x} \quad \rightarrow$$

$$y_p = y_{p_1} + y_{p_2} + y_{p_3}$$

$$y_{p_1} = Ax^2 + Bx + C,$$

$$y_{p_2} = E \cos 2x + F \sin 2x,$$

$$y_{p_3} = Gxe^{6x} + He^{6x}.$$

break it to three parts and find their particular solutions separately

$$y'' - 9y' + 14y = 3x^2$$

$$y'' - 9y' + 14y = -5 \sin 2x$$

$$y'' - 9y' + 14y = 8xe^{6x}$$

✓ We can find these particular solutions separately or simultaneously

Remark: No overlap between y_p and y_c

To define the particular solution y_p of a linear nonhomogeneous ODE:

- ✓ Identify all fundamental solutions of homogenous version (basis terms in the complementary solution y_c)
- ✓ If your trial y_p shares any term with y_c , multiply the trial by the smallest power of x to eliminate the match
(Avoid overlap).

Because:

The basis terms in y_c are the solutions of homogenous version and vanish the left side, and won't satisfy the nonhomogenous version

Example: Solve $y'' - 2y' - 3y = 4e^{3x}$

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 - 2r - 3 = 0 \rightarrow (r - 3)(r + 1) = 0$

e^{-x} and e^{3x} are fundamental solutions

$\rightarrow y_c = c_1 e^{-x} + c_2 e^{3x}$

Attention!

Define $y_p = Ae^{3x}$

$$\longrightarrow y_p' = 3Ae^{3x}$$

$$y_p'' = 9Ae^{3x}$$

Replace in $y'' - 2y' - 3y = 4e^{3x}$

$$\begin{aligned} 9Ae^{3x} - 2(3Ae^{3x}) - 3(Ae^{3x}) &= 4e^{3x} \\ &= 0 \end{aligned}$$

$\longrightarrow$ It doesn't give any solution

$y = Ae^{3x}$ can't be a solution of nonhomogenous version

Step2: Finding a particular solution of non-homogenous version

Define

$$y_p = Axe^{3x}$$



$$y_p' = Ae^{3x} + 3Axe^{3x}$$

$$y_p'' = 3Ae^{3x} + 3Ae^{3x} + 9Axe^{3x} = 6Ae^{3x} + 9Axe^{3x}$$

Replace in

$$y'' - y' + y = 5e^{2x}$$

$$(6Ae^{3x} + 9Axe^{3x}) - 2(Ae^{3x} + 3Axe^{3x}) - 3(Axe^{3x}) = 4e^{3x}$$

$$6Ae^{3x} + 9Axe^{3x} - 2Ae^{3x} - 6Axe^{3x} - 3Axe^{3x} = 4e^{3x}$$

$$4Ae^{3x} = 4e^{3x} \quad \longrightarrow \quad A = 1$$

$$\longrightarrow y_p = xe^{3x}$$

Step3: $y = y_c + y_p$

Example: Solve $y^{(4)} + y''' = 1 - x^2 e^{-x}$

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^4 + r^3 = 0 \rightarrow r^3(r + 1) = 0$

$e^{0x}, xe^{0x}, x^2e^{0x}$ and e^{-x} are fundamental solutions

 $y_c = c_1 + c_2x + c_3x^2 + c_4e^{-x}$

Step2: Finding a particular solution of non-homogenous version

$$g(x) = 1 - x^2 e^{-x}$$

→ $y_p = \overset{\downarrow}{A} + (Bx^2 + Cx + \overset{\downarrow}{D})e^{-x}$ However we may initiate with this guess,
since the constant and e^{-x} are seen in y_c

$$y_c = \overset{\downarrow}{c_1} + c_2 x + c_3 x^2 + \overset{\downarrow}{c_4} e^{-x}$$

so we change the trial y_p

Define $y_p = Ax^3 + x(Bx^2 + Cx + D)e^{-x}$